Reasoning Driven Captions To Assist Noise Robust Speech Emotion Recognition

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Problem

Speech Emotion Recognition (SER) degrade in presence of real-world noises.

Conventional: Solution

Integrating

1. Speech Enhancement (SE) [1]
2. Automatic Speech Recognition (ASR) [2]

Limitation

1. SE may introduce artifacts,
2. ASR model's performance degrade with noise.

Propose: Solution

We propose ***reasoning-driven captions*** as a new textual modality to assist SER under noisy conditions.

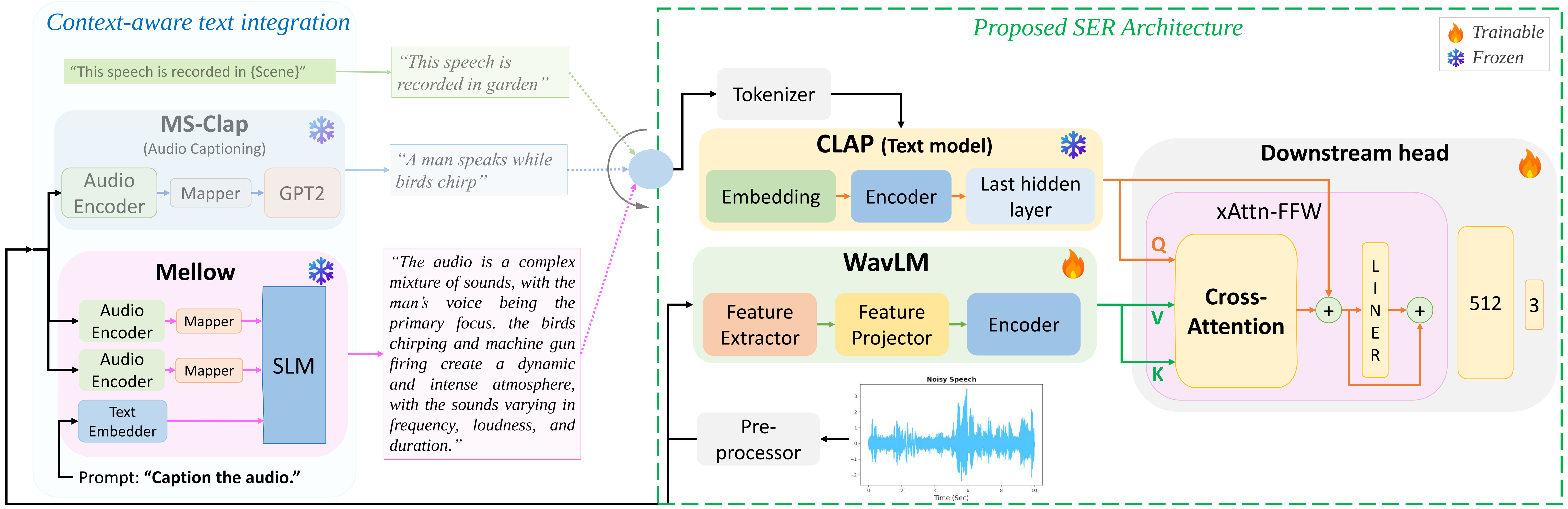


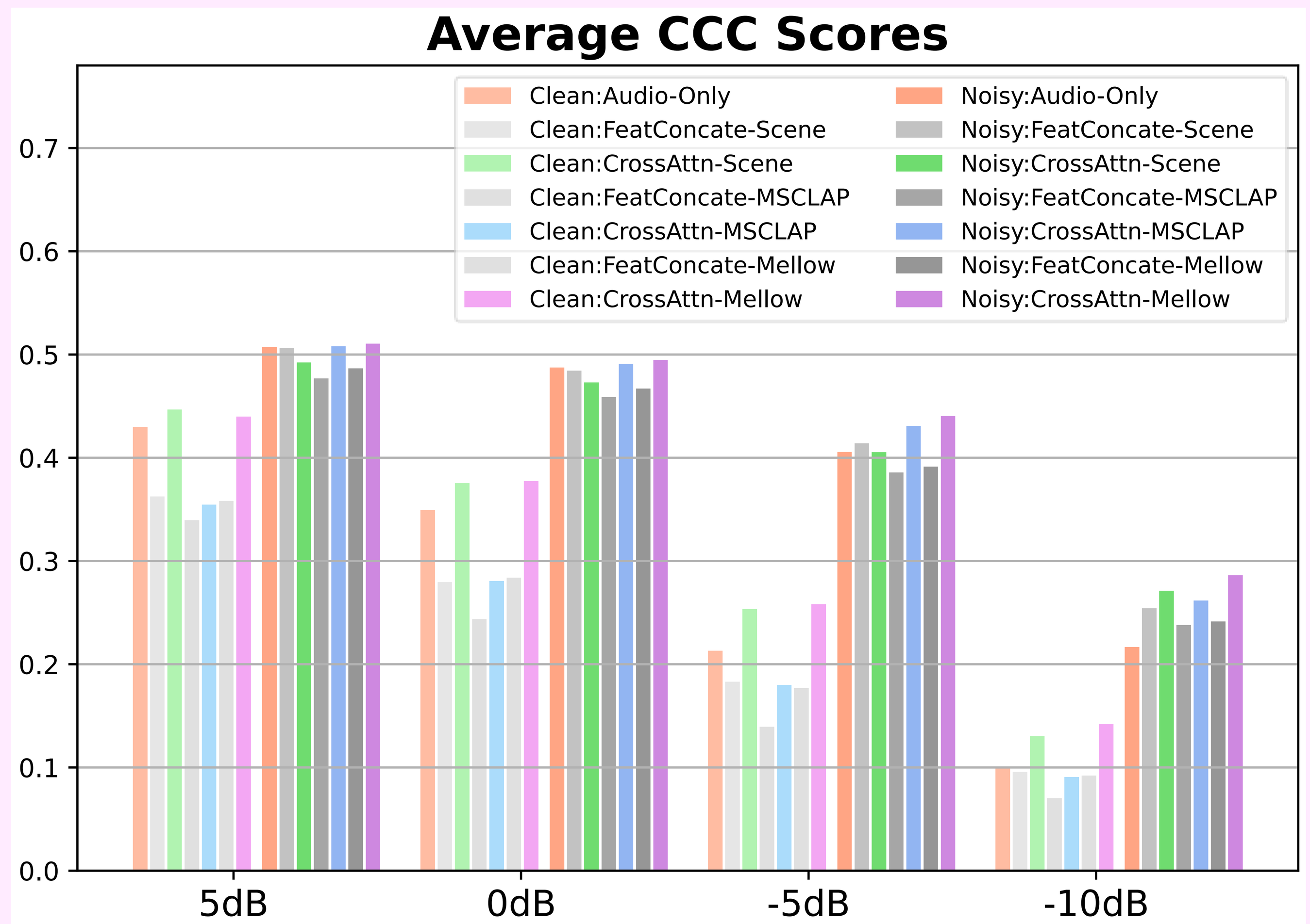
Fig. 1. Pipeline: (a) Context-aware text integration and (b) Proposed SER Architecture

Dataset

Set	Train	Validation	Test
Speech	MSP-Podcast (Train)	MSP-Podcast (Validation)	MSP-Podcast (Test)
Noise (Classes)	**20 Classes**: mall, restaurant, office, airport, station, city, park, street, traffic, home, kitchen, living room, bathroom, bedroom, metro, bus, car, construction site, pedestrian, beach		**6 Classes**: plaza, garden, school, tram, sea, boat
SNRs (dB)	{12.5, 7.5, 2.5}		{5, 0, −5, −10}

Pretrained Models

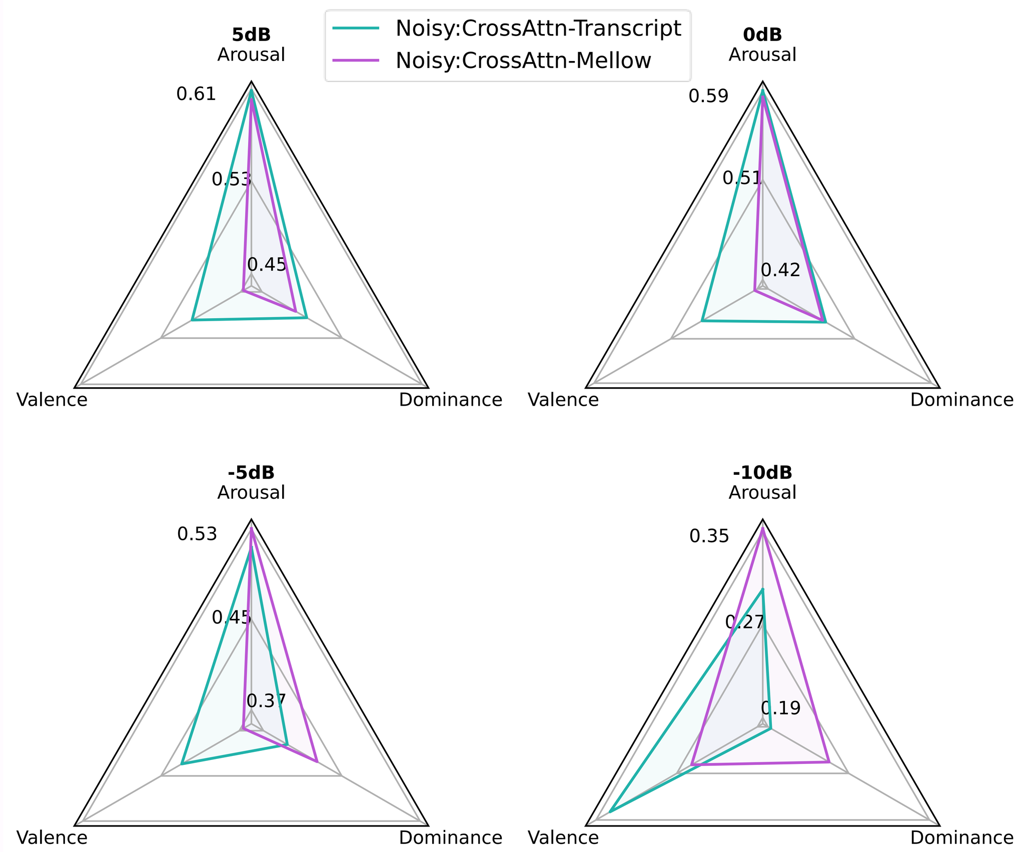
Models	Parameters [finetuned]
SER Architecture	
WavLM Base+	94.7 M [✔]
CLAP Text	125 M [✗]
Downstream Head	3.35 M [✔]
Context-aware text model	
MS-CLAP	227 M [✗]
Mellow	167 M [✗]
Total Trained Parameters	98 M



SNR	Score	Audio-only	Text-only		Baseline Audio+Text: Feature Concatenation			**Proposed** Audio+Text: Cross-Attention		
			Transcript	Mellow	Scene [3]	MS-CLAP	Mellow	Scene	MS-CLAP	**Mellow**
5dB	Arousal	0.5929	0.0912	0.0557	0.5911	0.5856	0.5899	0.5908	**0.6046**	0.6004
	Valence	0.4385	0.1410	0.0132	**0.4497**	0.3888	0.3939	0.4071	0.4272	0.4475
	Dominance	0.4909	0.0041	0.0073	0.4779	0.4564	0.4761	0.4791	**0.4922**	0.4837
0dB	Arousal	0.5736	0.0912	0.0552	0.5713	0.5673	0.5705	0.5594	**0.5852**	0.5847
	Valence	0.4122	0.1410	0.0119	0.4215	0.3684	0.3695	0.3957	0.4055	**0.4227**
	Dominance	0.4763	0.0041	0.0068	0.4604	0.4409	0.4611	0.4635	**0.4822**	0.4768
-5dB	Arousal	0.4808	0.0912	0.0492	0.5043	0.4844	0.4859	0.4743	0.5201	**0.5304**
	Valence	0.3460	0.1410	0.0036	0.3359	0.3110	0.3044	0.3408	0.3493	**0.3659**
	Dominance	0.3899	0.0041	0.0048	0.4017	0.3619	0.3840	0.4007	0.4232	**0.4248**
-10dB	Arousal	0.2484	0.0912	0.0415	0.3251	0.2984	0.2982	0.3195	0.3174	**0.3523**
	Valence	0.2155	0.1410	0.0035	0.1857	0.2086	0.2014	0.2371	0.2353	**0.2553**
	Dominance	0.1862	0.0041	0.0026	0.2518	0.2069	0.2242	**0.2568**	0.2323	0.2505

Analysis: Balanced score

Emotion	Happy 😊
Transcript	*"okay.and i have to look it up now because it's going to bug me if i don't."*
Mellow: Captions	*"The audio is loud and energetic, with a strong emphasis on the rhythmic thumping and pounding sounds. the adult female's voice is clear and prominent, but the other sounds are overpowering her."*



GitHub:

Source Code

References:

1. Chengxin Chen and Pengyuan Zhang, "Trnet: Two-level refinement network leveraging speech enhancement for noise robust speech emotion recognition," Applied Acoustics, 2024.
2. Yuanchao Li, Peter Bell, and Catherine Lai, "Speech emotion recognition with asr transcripts: a comprehensive study on word error rate and fusion techniques," in 2024 IEEE Spoken Language Technology Workshop (SLT), 2024, pp. 518–525.
3. Seong-Gyun Leem, Daniel Fulford, Jukka-Pekka Onnela, David Gard, and Carlos Busso, "Describe where you are: Improving noise-robustness for speech emotion recognition with text description of the environment," IEEE Transactions on Affective Computing, pp. 1–14, 2025.